

Priyanka U Ghodke

Electronics & Telecommunication Engineer

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SUMMARY

Motivated Electronics & Telecommunication Engineering student with hands-on experience in Research and Development Department working on low-noise analog circuit design, signal processing, and embedded systems, gained through internships at ISRO-SAC and NCRA-GMRT. Worked on SDD detector front-end readout design, focusing on low-noise amplification, signal conditioning, and space-grade system performance. Proficient in Analog Circuit Designing, MATLAB, LabView, Embedded C, and PSpice/OrCAD, with practical exposure to system modeling, Electronics Circuit Design, and control systems.

TECHNICAL SKILLS

- **Programming** : Embedded C, Python, C, Matlab Scripting, C++
- **Tools** : MATLAB/Simulink, LABView, MPLABX, Arduino, Orcad, Pspice, LTspice, Proteus, Vivado, Pcsimlab, VSCode, KiCad
- **Skills** : Analog Circuit Design, Electrical Circuits, Sensor Integration, Control Systems, Circuit Schematic Design and Analysis, Analog/Digital Electronics, Signal Processing, PCB Design, Electronics/Electrical Systems, I2C, SPI, UART, Transmitters & Signal Conditioning, Testing & Quality, Microcontrollers - STM32, ESP32, AVR or Atmel Boards, and PIC boards (by Microchip).

PROFESSIONAL EXPERIENCE

Indian Space Research Organisation (ISRO - SAC)

Sep 2025 - Mar 2026

Research Intern

Ahmedabad, Gujarat

- Designed and simulated a low-noise Op-Amp-based analog front-end readout for a Silicon Drift Detector using OrCAD PSpice, converting very small charge pulses into well-shaped output signals.
- Contributed to analog circuit design and testing under the Sensor Development Area (SEDA).

Tata Institute of Fundamental Research (NCRA - GMRT)

Jun 2025 - Aug 2025

Research Intern

Pune, Maharashtra

- Modeled wind turbulence data using FFT & Davenport Spectrum for antenna control systems.
- Developed transfer function models to improve control precision, performed system validation and analysis along.

Emertxe Technologies

Jan 2025 - Feb 2025

Technical Intern

(Hybrid)

Developed sequential control logic using PIC16F877A microcontroller and Embedded C.

PROJECTS

Design of Low-Noise SDD Detector Front-End Readout | *Pspice, Orcad*

Sep 2025

- Designing and simulating a low-noise Op-Amp-based front-end readout circuit for Silicon Drift Detector using PSpice/OrCAD under ISRO's Sensor Development Area.
- Implementing charge-sensitive amplification and signal conditioning techniques for accurate low-amplitude signal acquisition.
- Optimizing circuit performance by minimizing noise (thermal, shot, 1/f) and improving signal-to-noise ratio (SNR) for reliable space-grade operation.

Wind Data Analysis & Transfer Function Modeling | *Matlab*

Jun 2025

- Analyzed GMRT wind turbulence data of approx. 86,426 seconds to model antenna dynamics using FFT/Welch PSD and despiking techniques.
- Validated spectral characteristics against the Davenport spectrum and derived a 4th-order transfer function using the Tustin method.
- Quantified steady and dynamic wind loads, gaining hands-on experience in control systems, signal processing, and scientific instrumentation.

Solar Pesticide Sprayer Robot | *VS Code, Arduino IDE, C, Embedded C*

Mar 2025

- Developed a solar-powered pesticide spraying robot using microcontroller-based control system with feedback mechanism for efficient operation.
- Implemented actuators, relays, and DC motors for automated movement and pesticide spraying.
- Finalist at College of Engineering, Pune I2i Competition.

Microwave Oven Simulation using PIC16F877A | *Embedded C, MPLABX, Proteus, Picsimlab* **Jan 2025**

- Developing a microwave oven simulation using PIC16F877A microcontroller programmed in Embedded C on MPLABX.
- Simulating and testing the system in Proteus and Picsimlab to validate real-time embedded functionality.

Four Wheel Locomotive Robot | *AVR Board, Embedded C,C*

Nov 2024

- Moving four wheel robot based on AVR and Atmega based microcontroller
- Made multifunctional with pick and place mechanism for industrial automation like applications using servo motor.
- Controlled using Controller Console like gamepad.

Generation Of Electricity from waste materials | *Electrical Hardware, Solar based*

Mar 2024

- Designed a hybrid waste-to-energy system using anaerobic digestion, pyrolysis, and solar integration to generate electricity.
- Built a functional prototype converting waste materials into sustainable electrical power.
- Selected for **State-Level Technical DIPEX -2024 Competition**

ACHIEVEMENTS

- Selected for Internship at **ISRO-SAC**, Ahemdabad.
- Finalist at I2i Competition at COEP, Pune.
- Shortlisted **amongst top 50 students** from all over India for **SKA's First Training Workshop** at **Indian Institute of Technology**, Indore.
- Finalist at Dipex-State Level Technical Competition.

EDUCATION

Savitribai Phule Pune University - Govt. College Of Engg. & Research

2022 - 2026

Bachelor of Engineering in Electronics and Telecommunication - **CGPA - 7.51 (as of now)**

Pune, Maharashtra

Maharashtra State Board

2020 - 2021

H.S.C. (12th) - **Percentage - 82.33%**

Ahmednagar, Maharashtra

Maharashtra State Board

2018 - 2019

S.S.C. (10th) - **Percentage - 83.40%**

Ahmednagar, Maharashtra